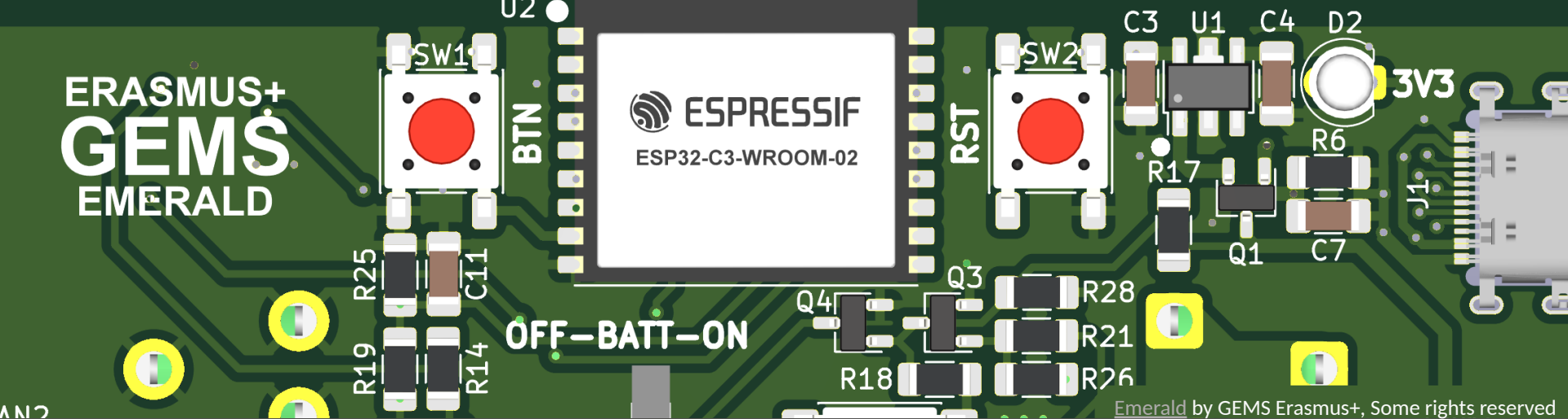


ERASMUS+
GEMS
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Wirelessly controlled robot – Control mode

Building Robots: System Integration

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About

1 Manual control

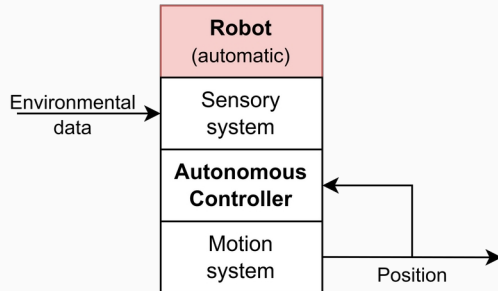
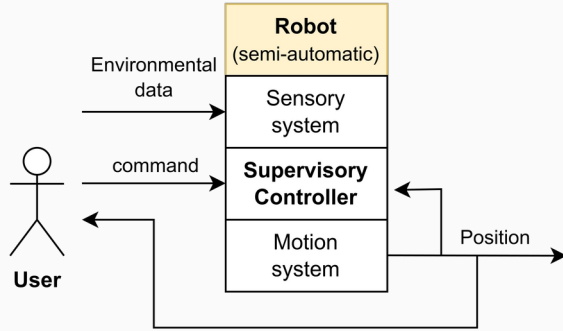
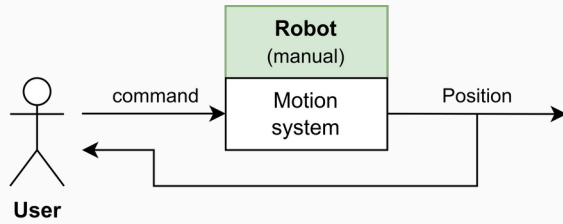
3 Safety

2 Automatic control

4 Extensibility

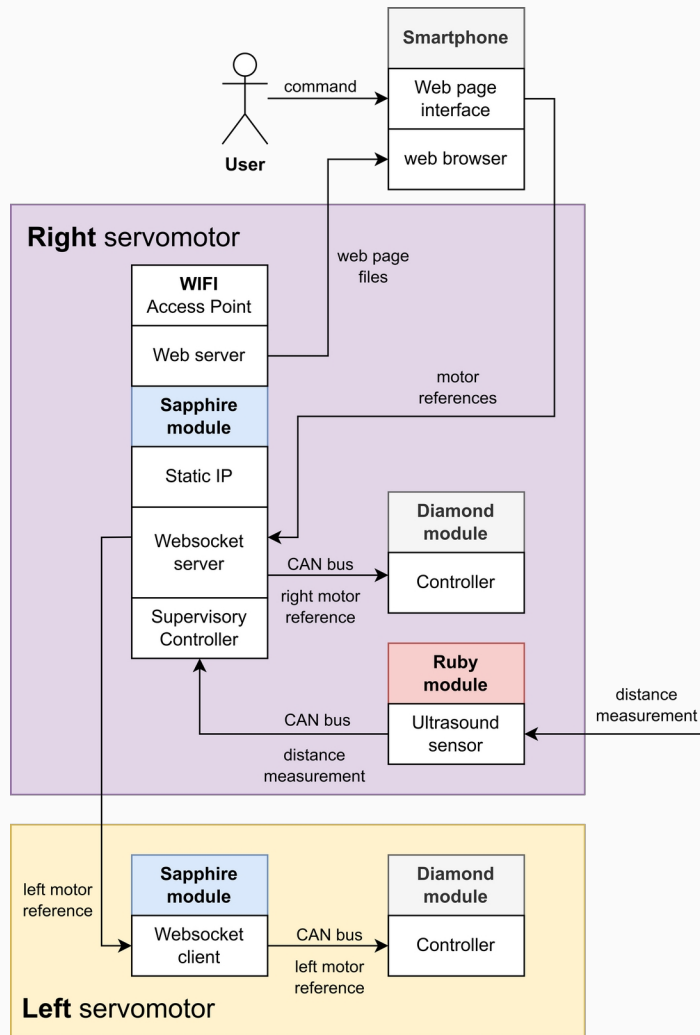
Modes of control





Robot control

- 1 **Manual: Operator directly controls the robot**
- 2 **Semi-automatic: Operator controls robot with assistance**
- 3 **Automatic: Robot controls itself (autonomous operation)**



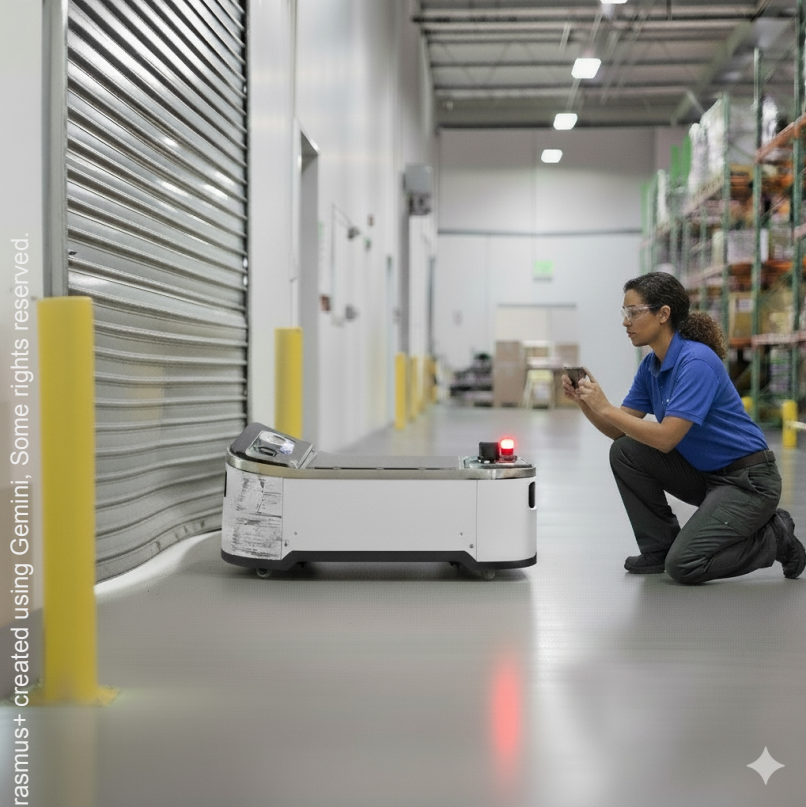
Semi-automatic mode

GEMS mobile robot example

- 1 Operator sends commands via wireless interface
- 2 Robot monitors its environment ultrasound sensor readings
- 3 The robot intervenes if a collision condition occurs → modifies operator's command

Safety





Requirements

- 1 Prevention of damage to:
people, equipment, self(robot)
- 2 Sources of damage: collision, pinch, crush, cut, component overload
- 3 Ability to stop the robot:
emergency stop button

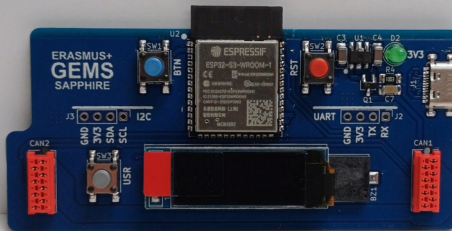
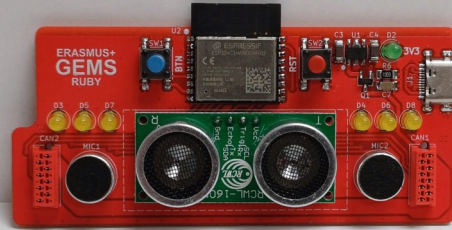
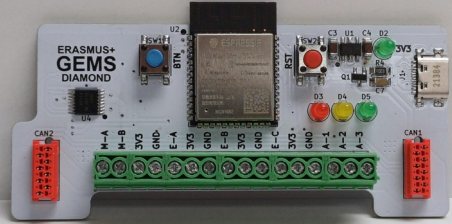
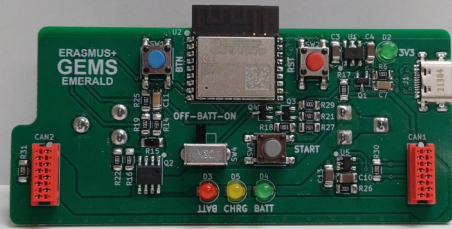


Implementation

- 1 Very predictable environment or monitoring (proximity or force)
- 2 Low-power operation:
low force, low speed, low mass
- 3 Harmless contact:
soft surfaces, rounded edges

Extensibility

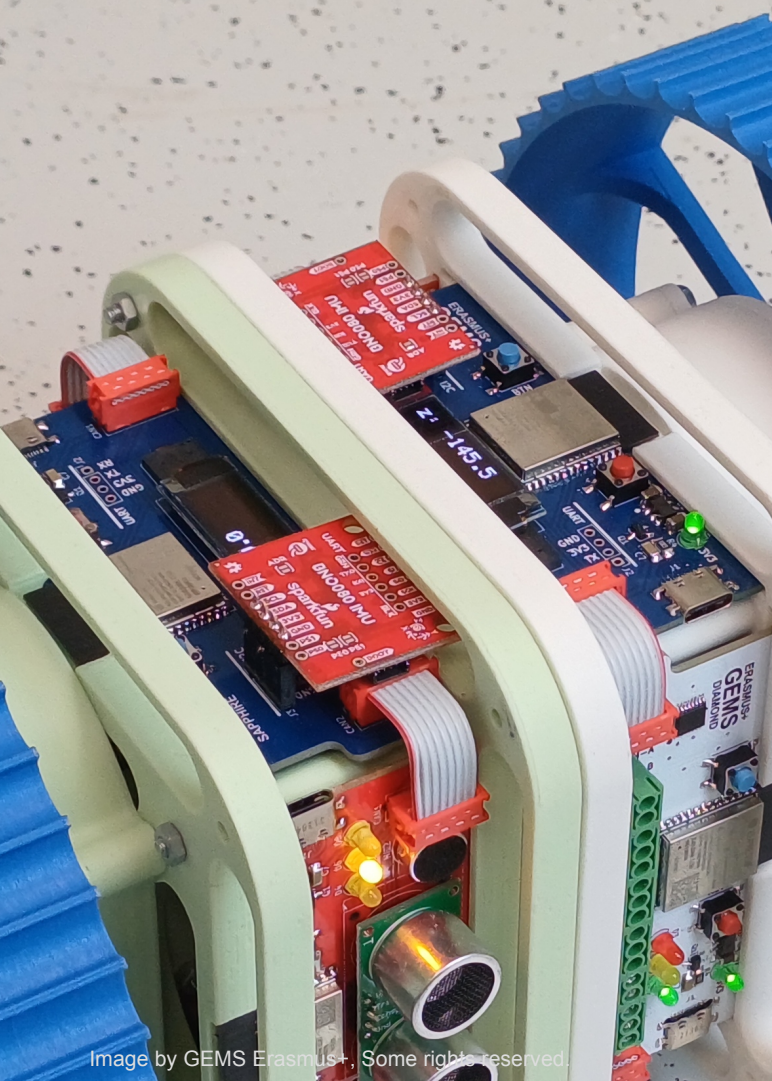




Example

GEMS educational hardware

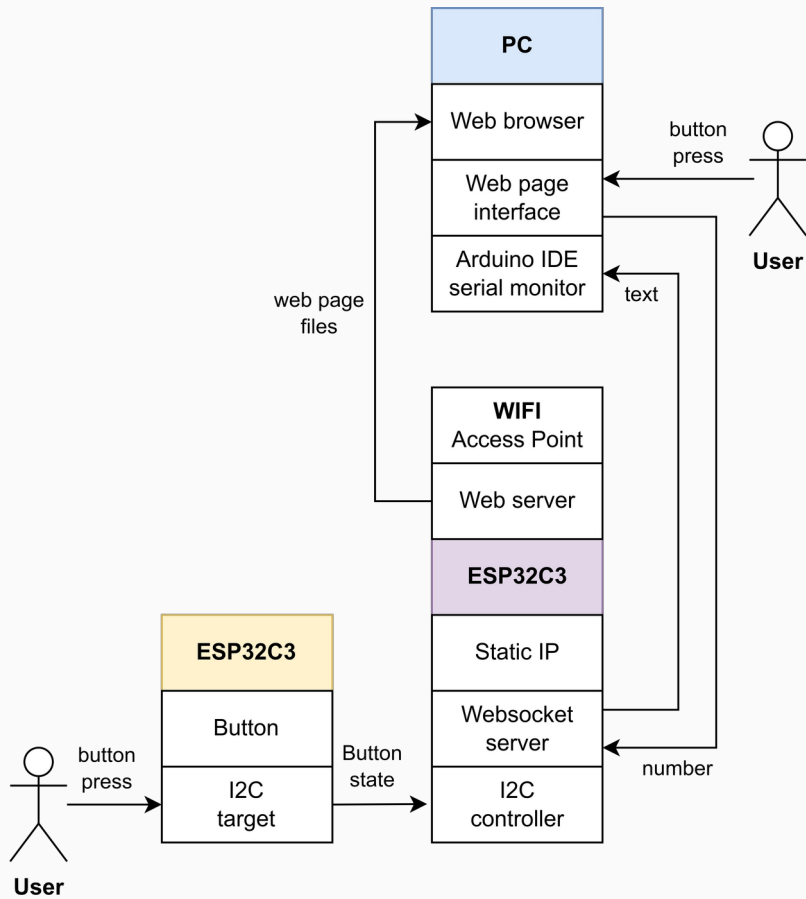
- 1 Modular design:
easy to replace/modify/reconfigure
- 2 Use of common elements:
well supported and documented
- 3 Multiple standard communication
interfaces: I2C, CAN, WiFi, BL,
UART, USB C



Implementation

- 1 Create options for upgrades
Documentation → usability
- 2 Black box – minimal information:
only about interface use
Open source – community support
- 3 Use of standard parts:
 - existing documentation
 - examples of use and tested

Task



Semi-automatic operation

Final system

- 1 2x ESP32C3/S3, browser wireless and wired
- 2 Safety condition: forward only if second button is also pressed
- 3 Communication between two microcontrollers

Conclusion

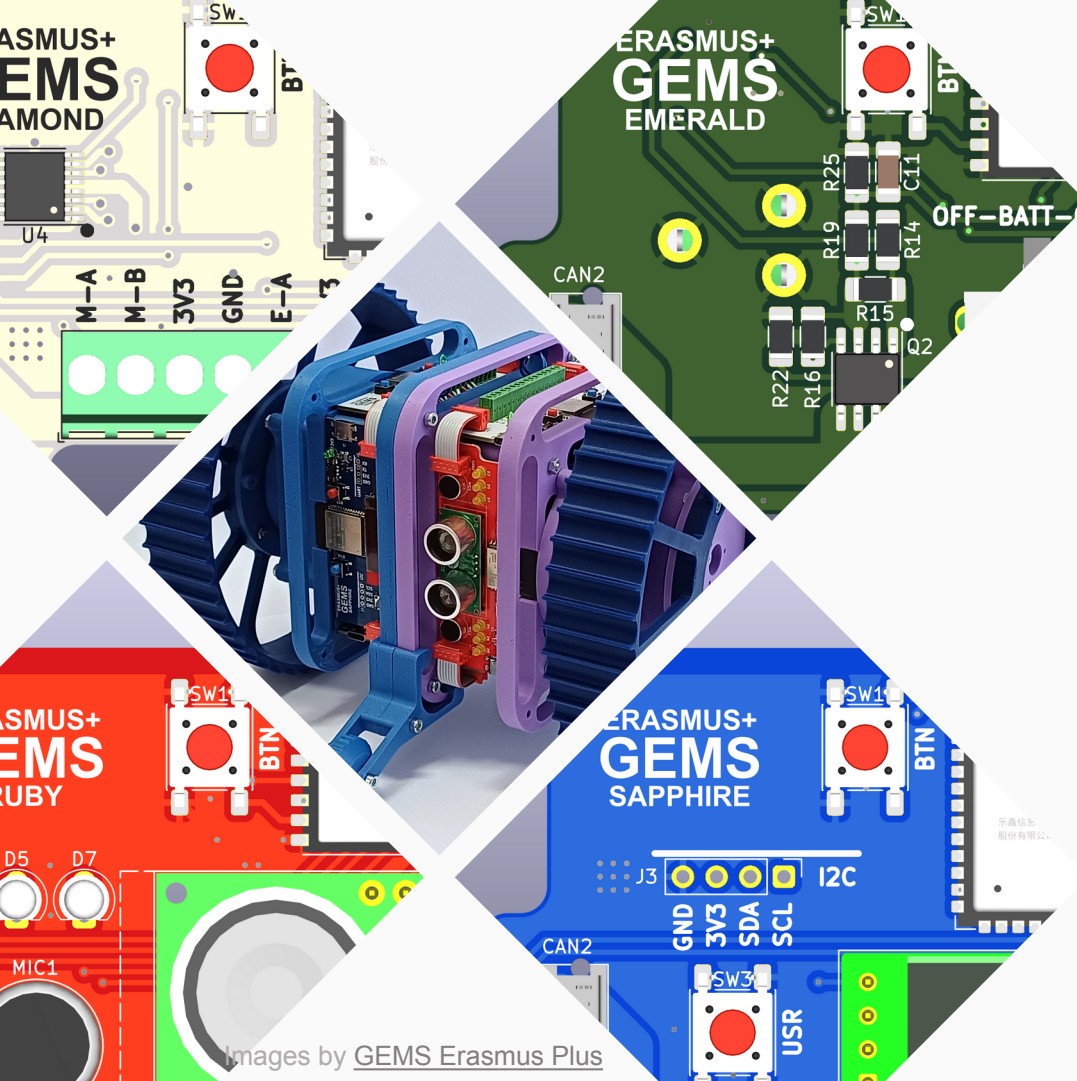




Create usefull robots

educational, industrial, scientific...

- 1 Robots are more usefull if they are safe with good user inteface
- 2 Tasks and specifications will change → design for future improvements
- 3 Create one robot → get knowledge to improve any robot



Images by [GEMS Erasmus Plus](#)

Thank you for watching!

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